

Appendix B: Capstone Projects

The book includes two major capstone projects that serve as comprehensive, self-paced take-home assignments. These projects consolidate your learning by applying the concepts to a realistic fleet and automotive services scenario.

Capstone 1: Global Fleet Intelligence

This project consolidates the foundations and prompt engineering skills covered in the first half of the book.

- **Exercise 1: Issue Classifier:** Applies a single well-structured prompt to produce reliable structured output.
- **Exercise 2: Work Order Parser:** Combines parsing, reasoning, and cost awareness on messy real-world input.
- **Exercise 3: Fleet Service Advisor:** An end-to-end task with an evaluation harness and prompt version comparison.

[Access Capstone 1 in the Companion Repo](#)

Capstone 2: Global Intelligent Fleet Assistant

This project focuses on the advanced concepts covered in the second half of the book, moving from single prompts to conversational, knowledge-grounded applications.

- **Exercise 1: Fleet Policy RAG System:** Implement a retrieval-augmented generation pipeline to answer driver policy questions.
- **Exercise 2: Guardrailed Maintenance Chatbot:** Build a multi-turn conversational agent with safety guardrails for both input and output.
- **Exercise 3: Maintenance Invoice Evaluator:** Build an automated evaluation harness to measure LLM accuracy against a golden dataset of messy invoices.

[Access Capstone 2 in the Companion Repo](#)